

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1351

North Huntington 138 kV
February 17, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1351. The Interconnection Customer proposes a 216.1 MW solar photovoltaic facility interconnecting at the North Huntington 138 kV substation in the Virginia area. The request was received on December 08, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 473 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$315,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Cloverdale / Hondo 138.0 kV L	Blue Ridge / Neely 138.0 kV	109.6	195	5.7
Vc Porterdaled Hydroelectric / Neely Substation 138.0 kV	Cloverdale Substation / Hondo 138.0 kV	100.6	277	8.6
Neely / Blueridge 138.0 kV	Cloverdale / Hondo 138.0 kV	100.2	295	8.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cloverdale / Hondo 138.0 kV L	Reconductor line	\$1,064,000	6% (\$61,000)
Vc Porterdaled Hydroelectric / Neely Substation 138.0 kV	Reconductor line	\$1,938,000	7% (\$133,000)
Neely / Blueridge 138.0 kV	Reconductor line	\$1,753,000	7% (\$121,000)
	Total allocated to project		\$315,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.